Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India

(Autonomous College Affiliated to University of Mumbai)

Mid Semester Examination

October 2025

Max. Marks: 30 | **Duration: one Hr**

Class: BE | **Semester: VII**

Course Code: CS402 | **Program: CE**

Name of the Course: Cryptography and System Security

Instruction:

(1) All questions are compulsory

(2) Draw neat diagrams and write equations wherever necessary

(3) Assume suitable data if necessary

Course Outcomes:At the end of the course student will be able to:

(1) CS402.1 Describe the different types of the cryptographic algorithms to secure information.

(2) CS402.2 Apply different cryptographic techniques to solve security-related problems.

(3) CS402.3 Create a message digest from data to authenticate authorized user

(4) CS402.4 Use system security practices

Q No.		Max. Marks	CO-BL
Q.1 (a)	A 16-round Feistel cipher uses 128-bit blocks and a 128-bit key. The key schedule sets $k_1, \ldots, k_8$ arbitrarily, then $k_9 = k_8$, $k_{10} = k_7$, $\ldots$, $k_{16} = k_1$. Given a ciphertext c and an encryption oracle, explain how to recover the plaintext m with a single oracle query. This shows that such a cipher is vulnerable to a chosen plaintext attack. (An encryption oracle is a device that outputs ciphertext for any given plaintext. Its internal working is unknown; you only learn by making queries and observing outputs.)	5	CO1, CO2

Q1 (b)	The Euclidean algorithm has been used for over 2000 years. In 1961, J. Stein introduced a new competitor called Stein's algorithm. It is as follows: Determine $\gcd(A, B)$ with $A, B \geq 1$. **STEP 1:** Set $A_1 = A,\ B_1 = B,\ C_1 = 1$ **STEP 2:** For $n > 1$, (1) If $A_n = B_n$, stop. $\gcd(A, B) = A_n C_n$. (2) If A_n and B_n are both even, set $$A_{n+1} = \frac{A_n}{2}, \quad B_{n+1} = \frac{B_n}{2}, \quad C_{n+1} = 2C_n.$$ (3) If A_n is even and B_n is odd, set $$A_{n+1} = \frac{A_n}{2}, \quad B_{n+1} = B_n, \quad C_{n+1} = C_n.$$ (4) If A_n is odd and B_n is even, set $$A_{n+1} = A_n, \quad B_{n+1} = \frac{B_n}{2}, \quad C_{n+1} = C_n.$$ (5) If A_n and B_n are both odd, set $$A_{n+1} = \lvert A_n - B_n \rvert, \quad B_{n+1} = \min(B_n, A_n), \quad C_{n+1} = C_n.$$ Continue to step $n + 1$. **i** Compute $\gcd(6150, 704)$ using both the Euclidean and Stein's algorithm. **ii** What is the apparent advantage of Stein's algorithm over the Euclidean algorithm?	5	CO1
Q.2	Encrypt the message "meet me" using the Hill cipher with the key $$\begin{pmatrix} 7 & 3 \\ 2 & 5 \end{pmatrix}.$$ Show the calculations for the corresponding decryption of the ciphertext to recover the original plaintext.	10	CO1, CO2

Q.3	Alice and Bob wish to agree on a secret key using a hybrid approach that combines **Diffie–Hellman key exchange** with **RSA encryption**. • Public Diffie–Hellman parameters: prime $q = 17$, primitive root $\alpha = 3$. • Alice's private DH key: $X_A = 5$. Bob's private DH key: $X_B = 4$. • Bob's RSA public key: $(e, n) = (7, 33)$, with corresponding private key $d = 3$. **i** Compute Alice's Diffie–Hellman public value $Y_A = \alpha^{X_A} \bmod q$. **ii** Show how Alice encrypts Y_A using Bob's RSA public key. Compute the ciphertext C_A. **iii** Show how Bob decrypts C_A with his RSA private key d to recover Y_A. **iv** Compute Bob's Diffie–Hellman public value $Y_B = \alpha^{X_B} \bmod q$, and then derive the shared secret key K from both Alice's and Bob's perspectives. Verify that the results agree. **v** Briefly discuss the benefit of encrypting Y_A with RSA, and state **two weaknesses** of this hybrid scheme. Suggest **one improvement** to make the protocol secure in practice.	10	CO2